
GRIDWISE

LLM-Assisted Smart Campus Energy Optimization Engine

Bridging Operator Natural Language with Deterministic LP Optimization & Enterprise Cyber Defense

Team: KUET_CODE_AND_CAFFEINE

Event: BUP Hackathon 2026

Backend Stack: FastAPI · HiGHS LP Solver (SciPy) · Pydantic v2

Frontend: Responsive SVG Console

Campus Energy Dispatch vs. Operator Intent

The Energy Dilemma Campuses face dynamic TOU peak tariffs (up to 34 BDT/kWh) and intermittent solar generation. Battery storage is crucial but degradation and capacity limits require precise scheduling.	The Operator Gap Operators communicate in natural language ('clean panels at noon', 'hold 80 kWh reserve'). Converting these to strict physical constraints manually causes delays and human errors.	LLM Pitfalls in Energy Raw LLMs hallucinate numbers, violate basic physical energy balance (supply != demand), and are vulnerable to prompt injection attacks if allowed to directly output schedules.
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Zero-Hallucination Deterministic Pipeline

1. Web Frontend	Interactive Console (SVG Energy Flow & Time Scrubber)
2. Security Middleware	Token-Bucket Rate Limiter · 1MB Payload Cap · IP Quarantine · Security Headers
3. Request Validation	Strict Pydantic v2 Models (24 Demand/Solar/Tariff arrays + Battery Physics)
4. LLM Interpreter	Adversarial Pattern Scanner · Start-Inclusive/End-Exclusive Normalization · 6 Schemas
5. HiGHS LP Solver	Min $\sum(\text{Grid}[h] * \text{Tariff}[h])$ · Hourly Energy Balance · Mandatory End-of-Day Neutrality $E[23]=E0$
6. Physics Replay Validator	Independent Replay Engine · Tolerance Check (0.01 kWh) · Recomputed Final Totals

Strict Semantic Normalization (6 Directive Types)

solar_reduction Usable solar multiplier (0.0 - 1.0). '80% solar reduction' → factor: 0.2	minimum_battery_reserve Battery energy floor E[h] >= min. 'Hold 80 kWh reserve' → min_energy: 80.0	no_charge_window Forced zero charge C[h] = 0. 'No charging from 2-4 PM' → [14, 15]
no_discharge_window Forced zero discharge D[h] = 0. 'Avoid discharge 6-8 PM' → [18, 19]	max_grid_window Grid import cap G[h] <= max. 'Limit grid intake to 155 kWh' → 155.0	no_op Informational or distractor notes. Applies = false, adjustment = null

Deterministic LP Formulation via HiGHS

Optimization Objective & Dynamics

- **Primary Objective:** $\min \sum (\text{Grid}[h] \times \text{Tariff}[h])$ for $h = 0..23$
- **Energy Balance:** $\text{Grid}[h] + \text{Solar}[h] + \text{Discharge}[h] = \text{Demand}[h] + \text{Charge}[h]$
- **Battery Transition:** $E[h] = E[h-1] + \text{Charge}[h] - \text{Discharge}[h]$
- **End-of-Day Neutrality:** $E[23] = E_{\text{initial}}$ (Mandatory)

Physical Constraints & Guarantees

- **Solar Curtailment:** $0 \leq \text{SolarUsed}[h] \leq \text{EffectiveSolar}[h]$
- **Battery Limits:** $\text{min_reserve}[h] \leq E[h] \leq \text{capacity_kwh}$
- **Power Bounds:** $0 \leq C[h] \leq \text{max_charge}$; $0 \leq D[h] \leq \text{max_discharge}$
- **Execution:** HiGHS LP Solver (<50ms execution, deterministic)

Enterprise Cyber Defense & Safe Fallback

Prompt Injection Defense

Scans operator notes for adversarial patterns ('ignore instructions', 'reveal keys') and blocks suspicious requests.

Token-Bucket Rate Limiter

Per-client IP sliding window rate limiting (60 req/min + burst capacity) prevents DoS.

LLM Circuit Breaker

CLOSED → OPEN → HALF_OPEN state machine isolates failing LLM APIs and activates safe fallback.

Sanitized Audit Logging

SOC2-compliant structured JSON logs automatically redact API keys, tokens, and passwords.

100% Automated Test Pass Rate & Official Benchmarks

38 / 38

Automated Tests Passed (100%)

10 / 10

Official BUP Challenge Cases Solved

< 50 ms

HiGHS Solver Execution Latency

0.01 kWh

Physical Replay Verification Tolerance

Test Suite Breakdown

- **test_validation.py**: 24-hour array shapes, non-negativity, battery physics.
- **test_directives.py**: 6 directive types, simultaneous constraint composition.
- **test_llm_guardrails.py**: Injection defense, circuit breaker transitions.
- **test_optimizer.py**: HiGHS global optimality, battery neutrality, windows.
- **test_schedule_validator.py**: Independent physics replay, corruption detection.
- **test_security.py**: Rate limiting token bucket, IP quarantine, JWT auth.
- **test_public_cases.py**: Full integration with all official challenge scenarios.

CONCLUSION

GridWise: Production-Ready Campus Energy Intelligence

Key Takeaways for Judges

- ✓ **Deterministic Trust:** Mathematical optimization guarantees the global lowest-cost schedule with zero hallucination risk.
- ✓ **Complete Web Architecture:** Live interactive frontend console mounted seamlessly on FastAPI backend.
- ✓ **Enterprise Cyber Defense:** Token-bucket rate limiting, prompt-injection defense, circuit breaker, and audit logging.
- ✓ **Containerized & Reproducible:** Multi-stage Docker setup with non-root security and automatic health checks.
- ✓ **Scalability:** Adaptable to university campuses, hospital microgrids, EV fleet depots, and smart districts.

Thank You! — [KUET_CODE_AND_CAFFEINE](#)